

saaf

Fix the English in your work messages with one keystroke.

saaf (साफ़) — Hindi/Urdu for clean. Which is the whole idea.

What this actually does

You are typing a message to a colleague, a manager, or a client — in Slack, in an email, in WhatsApp Web. The English comes out slightly wrong, the way it does when you are typing fast between two meetings. Normally you would open ChatGPT, paste the message, copy the answer back, and switch windows twice.

saaf removes that round trip. You select what you wrote and press one key.

WHAT YOU TYPED

hi sir, i has send the mail yesterday and i am
having fever so i will work from home

WHAT SAAF LEAVES BEHIND

**Hi sir, I sent the email yesterday and I have
a fever so I'll work from home.**

Nothing you write ever leaves this Mac. There is no account, no API key, no internet request. The language model runs on your own machine. This is the entire reason the tool exists instead of a browser tab — work messages stay on the laptop they were written on.

The whole thing in three steps

- 1 Select the text you just wrote. (Click and drag over it, or press **⌘ A** to select everything.)
- 2 Press **⌥ Option** + **P**. Wait about two seconds.
- 3 The corrected sentence has replaced your selection. Nothing else happens — no popup, no sound, no box. If you dislike the result, press **⌘ Z**, exactly like undoing anything else.

saaf is deliberately silent. The only thing that changes on screen is your text. If you would rather see what it edited and which rule you broke, that can be switched on — see *The menu bar icon* later in this guide — it is a checkbox, not a code change.

That is the complete user manual. Everything after this page is setup, detail, and what to do when something looks wrong.

Everything it can do

The whole feature list on one page, so you know what is there before the detail pages.

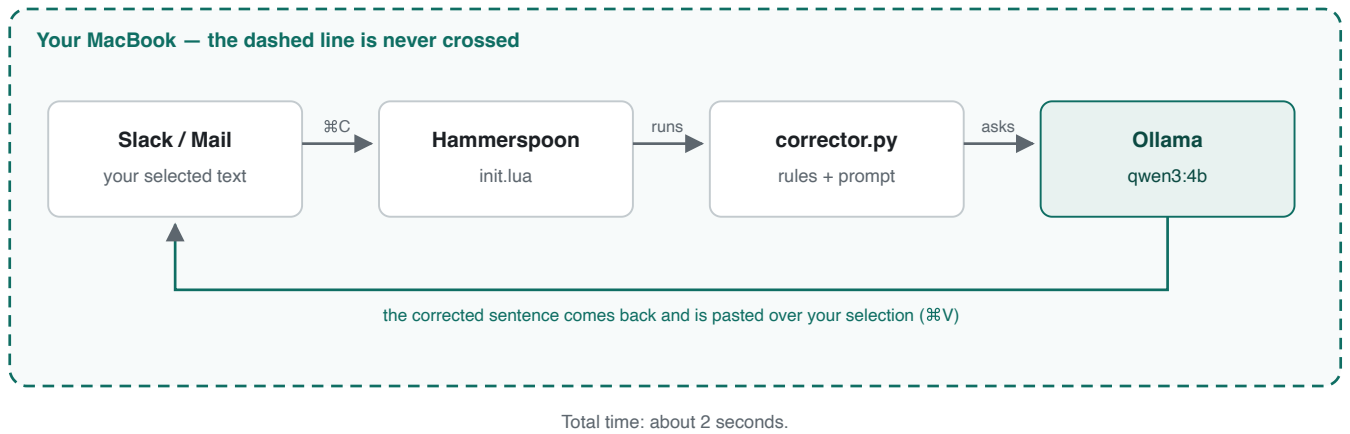
FEATURE	WHAT IT MEANS FOR YOU
One keystroke <code>⌘ Option</code> + <code>P</code>	Select text in any app, press it, and the corrected sentence replaces your selection in about a second and a half. <code>⌘ Z</code> undoes it like any other paste.
Nothing leaves your Mac	The language model runs on your own machine. No account, no API key, no internet request beyond <code>localhost</code> . This is the reason the tool exists.
Three audiences	Choose who the message is for — a colleague, a client, or Slack — and the register changes to match. See <i>Writing for someone specific</i> .
Silent by default	No popups over your work. The menu bar icon reports what is happening. You can turn on-screen detail on whenever you want it.
Menu bar control	On/off, audience, model status, today's count, start Ollama, start at login — all one click away, none of it requiring a file edit.
It refuses to guess	Phrases with no clear meaning, like " <i>do the needful</i> ", are flagged rather than silently interpreted. See <i>What it will not do to your writing</i> .
Learns you nothing, teaches you something	Every correction records which grammar rule you broke. <code>./corrector.py stats</code> turns that into a tally of your recurring mistakes.
Works on whole emails	Paragraph breaks, greetings and sign-offs survive intact. Select a four-paragraph draft and you get four paragraphs back.
Survives a restart	Starts at login, and keeps the model warm through the working day so you rarely wait.

Why two other programs have to be running

This is the part that confuses people first, so it is worth being clear. saaf is not a single app you launch. It is a small piece of glue that sits between two other programs, and **both of them must be running for the keystroke to do anything**.

PROGRAM	ITS JOB HERE	WHY SAAF CANNOT WORK WITHOUT IT
Ollama the brain	Runs the language model (qwen3:4b) on your Mac and does the actual correcting.	Without it there is nothing to correct your sentence. You get the message "Ollama is not running."
Hammerspoon the hands	Watches for <code>⌘ + P</code> , copies your selection, and pastes the result back.	Without it the keystroke does nothing at all — <code>⌘ + P</code> just types the character <code>π</code> .

How the pieces talk to each other



You do not have to keep track of this

That is what the **saaf icon in your menu bar** is for. You never have to remember whether either program is running, or open Terminal to find out — one glance at the top-right of your screen answers it:

ICON	MEANING	WHAT TO DO
◆	Ready	Nothing. Your next correction takes about a second and a half.
◇	Awake, model asleep	Nothing, but the next correction will take ~30 seconds. Click the icon → <i>Wake it now</i> if you would rather not wait mid-sentence.
◈	Working right now	Wait a moment. This is how you know a keystroke registered.
○	Switched off	You turned saaf off. Click the icon → <i>Turn saaf on</i> .
none	Hammerspoon is not running	This is the one thing the icon cannot report from the inside, so its absence is the signal. Launch Hammerspoon. It is set to start at login, so this should be rare.

Ollama not running is handled for you. Click the icon and choose *Start Ollama* — it launches the app and wakes the model, so the first correction afterwards is not also a slow one.

One-time setup

One command, then one click that only you can do. If saaf is already working on your Mac, skip this page.

1 Run the installer

Open **Terminal** (press `⌘ Space`, type *Terminal*, press Return), then drag the saaf folder into the window after typing `cd`, and run:

```
$ ./install.sh
```

It installs Ollama and Hammerspoon if they are missing, downloads the language model (about 2.5 GB, once), connects saaf to Hammerspoon, sets up the warm-up job, and tests that the engine works. Anything already done is skipped, so it is safe to run again if something looks wrong.

It will not overwrite an existing Hammerspoon setup. If you already use Hammerspoon, the installer adds one line to your config and backs the original up first, rather than replacing it.

2 Tick Hammerspoon in the Accessibility list

The installer opens the right settings pane for you at the end. Find **Hammerspoon** in the list and switch it on.

This is the one step that cannot be automated. macOS does not allow any program to grant itself Accessibility permission — that is a deliberate security rule, not a limitation of saaf. Without it the keystroke is accepted, nothing gets copied, and you are told "Select some text first" even though you did select something. The pane is **System Settings** → **Privacy & Security** → **Accessibility**.

3 Check for the ♦ in your menu bar

That icon is your confirmation everything loaded, and it is where all the settings live. If it is not there, quit and reopen Hammerspoon. Then select a badly written sentence anywhere and press `⌥ Option` + `P`.

To remove saaf

```
$ ./install.sh --uninstall
```

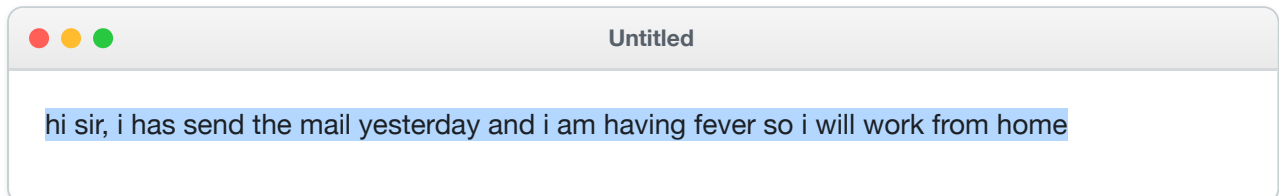
That unhooks saaf and restores your Hammerspoon config. Ollama, Hammerspoon, the model, and your `~/ .saaf/` correction history are deliberately left alone; the script prints the commands for removing those too if you want them gone.

Using it, step by step

The example below is TextEdit, but this works identically in Slack, Mail, WhatsApp Web, Notes, or any other app where you can select text. The corrected sentence shown is real output from the tool, not an approximation.

1 Write badly, then select it

Here is a genuinely mangled sentence. Select it — dragging over it works, or **⌘ A** selects everything in the box you are typing in.



Step 1. The blue highlight is the whole story — saaf only ever touches what is selected. It cannot see or change anything else on screen.

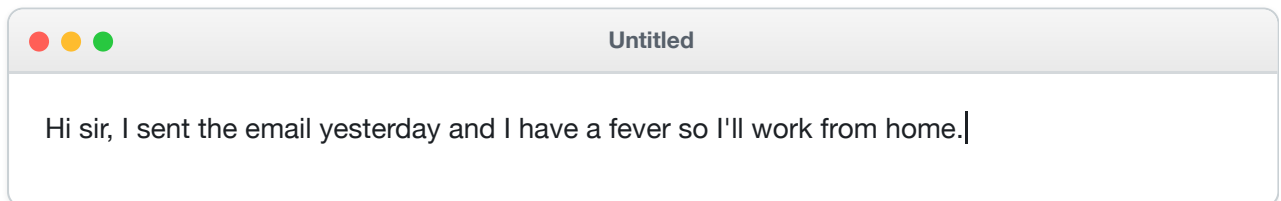
2 Press **⌘ + P** and wait about two seconds

Nothing appears over your work — no spinner, no popup. The only thing that moves is the menu bar icon, which changes to **◆** while it thinks and back to **◆** when it is done.

That icon is how you tell a working keystroke from a missed one. If it never changes to **◆**, the keystroke did not reach saaf. If it sits on **◆** for a long time, the model was asleep and is waking up — that takes about 30 seconds, once.

3 Your text is replaced. That is all.

The selection is gone and the corrected sentence is in its place, as though you had typed it that way. No box, no confirmation.



Step 3. Compare it with step 1 — four separate fixes, silently. The capital *H*, *i has send* to *I sent*, *mail* to *email*, and *i am having fever* to *I have a fever*.

Because there is no summary, **read the result before you send it.** It is accurate the overwhelming majority of the time, but you are now the only thing checking it.

Changed your mind? Press **⌘ Z**. saaf pastes like any other paste, so your app's normal undo puts your original wording straight back. Your clipboard is also left exactly as you found it — saaf borrows it for a moment and hands it back.

Writing for someone specific

The same message reads differently depending on who is going to read it. Click the menu bar icon, choose **Writing for**, and pick one. It stays until you change it, and the current choice is always shown in the menu and the tooltip.

CHOICE	WHAT CHANGES
Colleague the default	Normal work register. Contractions are fine, sentences stay short. This is what everything was tuned for.
Client or senior	Complete sentences and no contractions — <i>I will</i> rather than <i>I'll</i> . Precise and respectful, without becoming longer or adding flattery.
Slack / WhatsApp	Cut to the fewest words that stay polite. Roughly a third of the length. Facts, numbers and names always survive.

The same message, all three ways

Real output. The input is 52 words of the kind of thing you type in a hurry:

```
hi sir, i just wanted to quickly check in with you regarding the deployment
that we discussed about in yesterday meeting, i has completed the changes and
pushed it to staging, so whenever you get some time please can you kindly
review it and let me know your feedback on the same
```

Colleague 38 words	Hi sir, I just wanted to quickly check in about the deployment we discussed yesterday. I've completed the changes and pushed it to staging. When you get a chance, please review it and let me know your feedback.
Client 37 words	Hi sir, I just wanted to check in about the deployment we discussed yesterday. I completed the changes and pushed it to staging. Please review it when you have time and let me know your feedback.
Slack 18 words	Hi sir, I've completed the deployment changes and pushed to staging. Please review when you get a chance.

Slack mode takes about a second longer. It is produced by a different, more thorough pass — one that was already proven to shorten reliably. Three attempts at a faster method all gave inconsistent results (the same 52-word message came back at 50 words once and 28 the next time), so it uses the slower one that gives 16 to 18 words every time.

The setting sticks. If you leave it on *Client or senior* and forget, your next Slack message comes out formal. The menu bar tooltip tells you which is active without opening anything — hover over the icon.

What happens in those two seconds

You do not need any of this to use saaf. It is here because a tool that edits your writing and touches your clipboard should not be a black box.

TIME	WHAT HAPPENS	WHY IT IS DONE THAT WAY
0 ms	Hammerspoon catches <code>⌘P</code> before your app sees it. The icon turns  .	This is why the key never types a <code>π</code> . If you see one, Hammerspoon is not running.
0 ms	Your current clipboard is saved, then <code>⌘C</code> is pressed for you.	saaf has no way to read a selection directly — no app exposes one. Copying is the only route, which is why Accessibility permission is required.
250 ms	Your selected text is read, and your original clipboard is put straight back .	Restored immediately rather than at the end, so even a crash mid-correction cannot eat your clipboard. Slack and other Electron apps are slow to fill the clipboard; 250 ms covers them.
250 ms	<code>corrector.py</code> starts as a separate program and is handed your text.	Kept separate on purpose: the correction logic knows nothing about hotkeys or clipboards, so it can be tested on its own and run from Terminal.
~1.4 s	It asks Ollama, which runs the model on your Mac's GPU and returns a corrected sentence plus the grammar rules you broke.	The only slow part. Nothing leaves the machine; there is no network request beyond <code>localhost</code> .
~1.45 s	A fixed rule list runs over the answer — <code>kindly</code> → <code>please</code> , <code>revert back</code> → <code>reply</code> , and the rest.	Testing showed the model ignores these instructions about a third of the time even when told explicitly. Rules cannot forget, so anything expressible as a rule is a rule.
~1.5 s	The correction is put on the clipboard and <code>⌘V</code> is pressed.	Pasting rather than retyping is what makes <code>⌘Z</code> work — your app sees one ordinary paste, so one undo reverses it.
~1.65 s	Your clipboard is restored again, the icon returns to  , and the correction is appended to <code>~/saaf/mistakes.log</code> .	That log is what <i>My frequent mistakes...</i> counts. It never leaves your Mac and you can delete it any time.

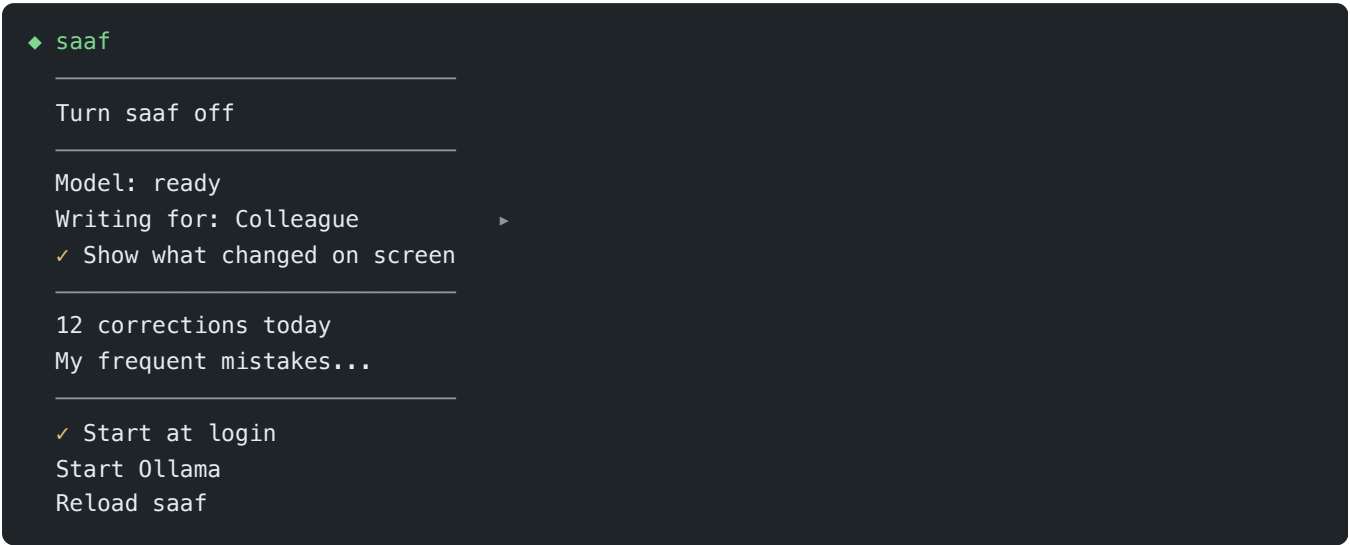
Why two layers instead of just asking the model

The model handles grammar, which needs judgement. A fixed rule list handles office filler and a short set of recurring mistakes, which do not. Splitting it that way is what lets saaf use a small model that answers in about a second instead of a large one that answers in five — the small model only has to get the grammar right, because everything else is already guaranteed.

It is also why one specific thing is *never* auto-fixed. "*Do the needful*" has no safe replacement — it could mean review, approve, deploy, or pay — so instead of guessing, saaf flags it and leaves the sentence to you.

The menu bar icon

Everything you can change about saaf is one click away. Nothing here requires editing a file or restarting anything.



Turn saaf off	Stops corrections without quitting anything.  does nothing while off — it will not start typing a π at you either. Useful when you are writing code or in a language saaf should not touch.
Model: ready / asleep	Whether the next correction is fast or slow. When asleep, a <i>Wake it now</i> entry appears underneath.
Writing for	Who the message is going to: Colleague, Client or senior, or Slack / WhatsApp. The active choice is shown on this row, so you can see it without opening the submenu. See <i>Writing for someone specific</i> .
Show what changed on screen	Off by default. Tick it and every correction shows what it edited, which grammar rule you broke, and any warnings. Takes effect immediately — tick it for one important email and untick it after.
My frequent mistakes...	Opens the same tally as <code>./corrector.py stats</code> in a text window.
Start at login	Already on. This is what makes saaf survive a restart — without it, a reboot leaves you with no icon and a dead hotkey.
Start Ollama	Launches Ollama and wakes the model. This is the "I don't want to think about it" button.

What "show what changed" looks like

With that box ticked, a dark panel appears in the centre of the screen while it works, and a second one afterwards showing up to three kinds of information. Here is one with all three at once:

hi → Hi i has send → I sent mail yesterday and pls →
email yesterday. Please the needful → what's needed.
subject-verb agreement
comma splices
"do the needful" does not say what you want. Name the
action: review it, approve it, deploy it.
⌘Z to revert

The green line — the edits. Literally what was swapped, old on the left, new on the right. Skim it to confirm your meaning survived.

The plain lines — the grammar rules you broke. Named, so they are learnable rather than merely fixed. These get counted; see the next page.

The amber line — a warning. Something saaf is telling you it could not safely decide for you. This is the important one.

The grey line — the reminder that ⌘Z undoes the whole thing.

Why it warns instead of trusting the fix

"*Do the needful*" could mean review it, approve it, deploy it, or pay it. Nothing in your sentence says which. In the example above the model guessed "*what's needed*" — which is no more specific than what you wrote, and could still send a colleague off in the wrong direction.

So the warning does not depend on the model behaving well. **saaf checks the phrase against what you typed, not against what came back** — which means it fires either way, whether the model left the phrase alone or quietly paraphrased it. The fix it wants from you is to name the action yourself. **This is the one place saaf teaches instead of correcting, and it is deliberate.**

This is what running silent actually costs you. The warning still gets calculated on every correction — it just has nowhere to appear. A vague instruction gets silently paraphrased into something equally vague and nothing points it out. If you write "*do the needful*", "*revert back*", or similar office shorthand often, this is the single best reason to tick *Show what changed on screen*. Otherwise, check the sentence in Terminal when it matters:
`./corrector.py --fast "your text"` prints the warning every time.

Phrases that *do* have a safe replacement are simply fixed without comment — for example `kindly` becomes `please`, `revert back to` becomes `reply to`, and `pls`, `thx`, `asap` are always spelled out in full.

Things saaf will never do

Never adds a greeting or sign-off	No "Dear", no "Regards", no "Looking forward to hearing from you."
Never invents a fact, name, date, or number	If you did not write it, it will not appear.
Never adds confidence you did not have	If you wrote "i think", the doubt stays.
Never turns a one-line message into a paragraph	A WhatsApp line stays a line.
Never makes it sound like AI wrote it	No "Additionally", "Furthermore", "Moreover"; no em dashes. The goal is correct English that still sounds like you in a hurry — not a press release.

What it will not do to your writing

A tool that rewrites your messages has to be trustworthy about what it leaves alone. Each of these is a real failure that was caught while building, and each is now prevented by a fixed rule rather than by asking the model nicely — because the model ignores that kind of instruction often enough to matter.

IT WILL NEVER...	WHY THIS RULE EXISTS
Attach a currency you did not write	Given "invoice for 45000", the model returned \$45,000 in eight runs out of eight. A bare number written in Mumbai is not dollars, and a wrong currency in an invoice is a real-money mistake. If you type \$45,000 or Rs 45000 , that is exactly what you get back.
Misspell your sign-off	A message ending "thanks" came back as "tanks" in five runs out of six. Greetings and sign-offs are your voice, not grammar, so they are now copied through and repaired if the model mangles them.
Flatten your paragraphs	A seven-line email used to come back as a single line with the sign-off missing. Blank lines are structure you chose. Four paragraphs in, four paragraphs out.
Guess what you meant	"Do the needful" could mean review, approve, deploy or pay. Rather than pick one and send a colleague the wrong instruction, saaf leaves the phrase alone and warns you to name the action yourself.
Drop a number or a name	Even in Slack mode, where the whole point is cutting words. Length can go; facts cannot.

All of these are tested, not promised. Each one has a case in the test suite that fails if the behaviour ever comes back. Run `python3 tests/bench.py qwen3:4b` to check them yourself — it prints every failure with the offending output.

The Terminal version (optional)

The keystroke gives you one corrected sentence, which is what you want almost every time. Occasionally you want choices, or you want to see which mistakes you keep making. Those live in Terminal. Nothing here is required to use saaf day to day.

Three phrasings instead of one

Run it with your sentence in quotes and you get a minimal fix, a natural rewrite, and a short version:

```
$ ./corrector.py "i am having 4 years experience and i discussed about it since 2 years"

1. I've got 4 years of experience and I've been discussing it for 2 years
2. I have 4 years of experience and I discussed it for 2 years
3. I've got 4 years of experience and 2 years of discussion

rules you broke: 'having' for experience | since

3.14s · qwen3:4b · nothing left this Mac
```

Option 2 is the one the hotkey would have given you. Add `--fast` to get just that one, in about half the time.

Writing for a particular reader

The same three audiences as the menu, without changing the menu setting:

```
$ ./corrector.py --formal "i has send the mail and i will share report by monday"

I have sent the email and I will share the report by Monday.

$ ./corrector.py --brief "hi sir, i just wanted to quickly check in regarding the
                        deployment, i has completed the changes and pushed to staging"

Hi sir, I've completed the deployment changes and pushed to staging.
```

Either flag gives you one answer rather than three, because an audience replaces the minimal / natural / short choice rather than stacking on top of it.

The office phrases that are always replaced

Some corrections do not involve the model at all. A fixed list of office filler has a safe replacement defined for it, and those swaps are applied every single time, in no measurable time:

```
$ ./corrector.py --fast "kindly revert back on this asap"

Please reply to this as soon as possible

1.2s · qwen3:4b · nothing left this Mac
```

`kindly` → `please`, `revert back` → `reply`, `asap` → `as soon as possible`. Because these are rules rather than requests to a model, they cannot be forgotten or ignored — run that command three times and

you get the identical sentence three times. The same applies to `pls`, `thx`, `u`, and `discussed about` → `discussed`.

Which mistakes you actually keep making

Every correction is logged to `~/ .saaf/mistakes.log` on your machine. This turns that log into a tally, so the point stops being "patch this message" and starts being "stop making this mistake":

```
$ ./corrector.py stats

39 messages corrected, first on 2026-08-12

Your most frequent mistakes:

present perfect with a past time word    19 #####
missing article (a / an / the)           9  #####
singular / plural                         6  #####
revert back                              4  ####
asap                                       4  ####
did not get (verb form)                  1  #
```

Read that as a study list. Nineteen of those thirty-nine messages used a present perfect with a past time word — "*I have sent it yesterday*" instead of "*I sent it yesterday*". That is a single rule, and learning it would remove almost half the corrections saaf has ever had to make for you.

The log is yours alone. It is a plain text file at `~/ .saaf/mistakes.log`, one line per correction, on your machine. Delete it whenever you like — saaf simply starts counting again.

When something looks wrong

Because saaf runs silent, the errors below arrive as ordinary macOS notifications in the top-right corner — not as centre-screen boxes. Anything in a `code style` below is the exact wording you will see.

Start with the icon. Almost every problem is identified by what the menu bar shows before you check anything else. **No icon** → Hammerspoon is not running. `○` → you turned saaf off. `◇` → it will work but slowly. **◆ that never changes to ◆ when you press `⌘P`** → the keystroke is not reaching saaf.

WHAT YOU SEE	WHAT IT MEANS AND WHAT TO DO
No saaf icon in the menu bar	Hammerspoon is not running — nothing else can be wrong yet, because the icon <i>is</i> Hammerspoon. Launch it from Applications. It is set to start at login, so if this keeps happening check that <i>Start at login</i> is still ticked.
Icon shows <code>○</code>	saaf is switched off. Click the icon → <i>Turn saaf on</i> .
Text unchanged, and a <code>π</code> appeared	The <code>π</code> means Hammerspoon never intercepted the key, so macOS typed the character instead. If the icon is present, the config loaded but the hotkey did not bind — click the icon → <i>Reload saaf</i> .
Icon flicked to <code>◆</code> and back, but the text did not change	It ran and found nothing to fix — your sentence was already correct. saaf leaves correct text alone rather than rewriting it for the sake of it.
<code>Ollama is not running.</code>	The brain is not awake. Click the saaf icon → <i>Start Ollama</i> . It launches the app and wakes the model for you.
<code>Select some text first. but you did select text</code>	Hammerspoon is missing Accessibility permission, so its copy did nothing. Grant it in System Settings → Privacy & Security → Accessibility, then reload the config.
A correction takes ~30 seconds icon was showing <code>◇</code>	The model was asleep and is being read off disk. Working as intended, just slow once — every correction after it is fast. Avoid it by clicking <i>Wake it now</i> whenever you notice <code>◇</code> , and by installing the background job from step 4 of the setup page.
<code>Model did not finish loading in 90s.</code>	The load stalled. Run <code>ollama run qwen3:4b</code> once in Terminal, wait for a prompt, then try again.
<code>Model took longer than 25s.</code>	Your selection was long. Select a paragraph at a time rather than a whole email.
<code>Text is N characters. Select less than 4000.</code>	Too much at once, for the same reason. Do it in pieces.
The correction is wrong or lost my meaning	Press <code>⌘Z</code> . Then, if the sentence was ambiguous, rewrite the unclear part yourself — that is usually the real fix. Silent mode means you are the only reviewer, so re-read anything important before sending.
I want to know what it changed	Click the icon and tick <i>Show what changed on screen</i> . Or check a sentence in Terminal before using it: <code>./corrector.py --fast "your text"</code> .

The one thing that fixes most things

Click the saaf icon and choose **Reload saaf**. That re-reads every saaf file and rebuilds the hotkey — worth trying before anything else, and the fix after you edit any of them. From Terminal the equivalent is `hs -c 'hs.reload()'`.

If the icon itself is missing, there is nothing to click: launch Hammerspoon from Applications. If it is misbehaving in some other way, quit and reopen it.

A useful habit: run `./corrector.py stats` once a week. The hotkey fixes today's message; the tally is the part that eventually makes the hotkey unnecessary.

saaf — local English correction for work messages · runs on qwen3:4b via Ollama, driven by Hammerspoon · **nothing you write leaves this Mac**